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4E

$$A_2 = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$$

$$R_1: \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 5 & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$R_2: \left[\begin{array}{ccc|ccc} 1 & 1/2 & -1/2 & 1/2 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 5 & 2 & -3 & 0 & 0 & 1 \end{array} \right] \text{ by } (1/2)R_1$$

$$R_3: \left[\begin{array}{ccc|ccc} 1 & 1/2 & -1/2 & 1/2 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & -1/2 & 1/2 & -5/2 & 0 & 1 \end{array} \right] \text{ by } R_3 - 5R_1$$

$$R_1 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -1/2 & 1/2 & 1/4 & 0 \\ 0 & 1 & 1/2 & 0 & 1/2 & 0 \\ 0 & -1/2 & 1/2 & -5/2 & 0 & 1 \end{array} \right] \begin{array}{l} \text{by} \\ \\ (1/2) R_2 \end{array}$$

$$R_2 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 3/4 & 1/2 & -1/4 & 0 \\ 0 & 1 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & -1/4 & -5/2 & 1/4 & 1 \end{array} \right] \begin{array}{l} \text{by } R_1 - (1/2)R_2 \\ \text{by } R_3 + (1/2)R_2 \\ \end{array}$$

$$R_3 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -3/4 & 1/2 & -1/4 & 0 \\ 0 & 1 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right] \text{by } (-4)R_3$$

$$R_1 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 1/2 & 0 \\ 0 & 1 & 0 & -5 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & -4 \end{array} \right] \begin{array}{l} \text{by } R_1 + (3/4)R_3 \\ \text{by } R_2 - (1/4)R_3 \\ R_3 \end{array}$$

$$\therefore A^{-1} = \left[\begin{array}{ccc} 8 & 1/2 & 0 \\ -5 & 0 & 0 \\ 0 & -1 & -4 \end{array} \right]$$